

5. (a) (i) Let T be the tension.

$$2T \cos 75^\circ = 60$$

$$T = 115.9 \text{ N}$$

1M

1A 2

- (ii) energy stored in the string = k.e. of arrow
 $= \frac{1}{2}(0.2)(45)^2$
 $= 202.5 \text{ J}$

1M

1A 2

- (b) (i) $d = v \cos 20^\circ t$
 $60 = 45 \cos 20^\circ t$
 $t = 1.42 \text{ s}$

1M

1A 2

- (ii) $h = 25 - \frac{1}{2}gt^2$
 $= 25 - \frac{1}{2}(9.81)(1.42)^2$
 $= 15.1 \text{ m [or } h = 14.9 \text{ m]}$

1M

1A 2